University of North Carolina at Pembroke

Department of Physics and Chemistry

Biomimetic Atmospheric Water Harvester/

Nature-Inspired Surface Designs for Sustainable Water Collection

---

Undergraduate Research Project

Emmanuella Okpara

3+2 Mechanical Engineering

Faculty Mentor

Mason Parkes

July – August 2026

Section	Purpose
* Abstract	Problems, approach, literature-based methods, synthesis, conceptual design, significance.
* Introduction	Freshwater scarcity, atmospheric water, passive collection, biomimetic motivation.
* Background	Fog/dew collection, environmental conditions, capture/transport/collection stages.
* Surface Wettability & Droplet Physics	Young, Wenzel, Cassie-Baxter, hysteresis, Laplace pressure, capillarity, gravity.
* Biological Models	Namib beetle, cactus spine, spider silk, mango leaf and supporting models.
* Comparative Literature Analysis	Systematic comparison of Sources 1-10.
* Experimental Evidence	Published performance values with their test conditions and limitations.
* Research Gap	Durability, scalability, standardized testing, integration, field relevance.
* Proposed Conceptual Design	CAD architecture, component rationale, mechanism chain, design variables.
* Future Validation	Controls, variables, metrics, hypothesis.
* Engineering Significance / Limitations	Mechanical engineering relevance and claim boundaries.
*Conclusion	Answer the research question using literature-supported evidence.
*References	All ten sources in the required citation style.

1.Abstract

Atmospheric water harvesting (AWH) is an approach for obtaining freshwater from moisture present in the atmosphere. The reviewed literature describes several routes, including fog interception, dew condensation, and moisture sorption. Within passive AWH, surface engineering is particularly important because the geometry and chemistry of a collector influence whether incoming water is captured, retained, merged, transported, and finally removed from the active surface.

A central problem is that capture and removal can require different interfacial behaviors. Hydrophilic surfaces can promote water interaction and deposition, but excessive adhesion or spreading can cause films and droplet pinning. Hydrophobic or low-adhesion surfaces can support droplet growth and shedding, but very low adhesion is not automatically ideal for initial capture. The literature therefore increasingly supports spatially patterned wettability rather than a single uniform surface property.

This research synthesizes ten peer-reviewed sources addressing biomimetic water-harvesting materials, passive atmospheric water harvesting, cactus-inspired fog collection, patterned wettability, bioinspired fog-collection structures, mango-leaf replication, and 3D-printed dual-wettability surfaces. The principal biological models are the Namib Desert beetle, cactus spines, spider silk, mango leaves, and shorebird beaks. Together they provide complementary solutions to localized capture, curvature-driven transport, periodic droplet accumulation, hierarchical drainage, and capillary movement.

Across literature, the strongest recurring principle is structure-wettability synergy. Tapered structures can generate curvature-dependent pressure differences, microgrooves can constrain droplet movement, wettability gradients can create energetic directionality, and larger channels can use gravity to deliver water to a reservoir. Zhang and Guo (2020) describe the broad sequence as capture, transportation, and storage, while Pan et al. 2026) and Dong et al. (2025) organize the process into capture/nucleation, coalescence, directional transport, and collection.

This research is intentionally literature-based. No new prototype, laboratory measurement, or collection-rate experiment is claimed. Published values are used only as evidence from the original studies. The engineered collector proposed later in the paper is therefore a conceptual design hypothesis rather than an experimentally validated device.

CENTRAL THESIS

Passive atmospheric water harvesting can be improved by coordinating heterogeneous wettability with hierarchical geometry so that different regions specialize in capture, droplet growth, directional transport, surface renewal, and collection.

2. Introduction and Research Context

2.1 Freshwater scarcity and atmospheric water

Freshwater availability is uneven across geographic regions and seasons. Arid and semi-arid areas can experience limited rainfall, groundwater stress, or expensive access to centralized infrastructure. These conditions motivate research into supplementary water sources, including water present in atmospheric fog and vapor. Liu, Beysens, and Bourouina (2022) place passive AWH within a broader group of approaches for obtaining water from air and emphasize that atmospheric water is geographically widespread but environmentally variable.

Atmospheric water should not be described as an unlimited reservoir of immediately recoverable liquid. The amount that can be collected depends on humidity, temperature, fog availability, airflow, droplet size, surface condition, and the collector's ability to move water after capture. A strong design must therefore address both the surface and the environment in which it operates.

2.2 Definitions: fog, dew, and sorption

Fog harvesting captures liquid droplets that already exist in fog-laden air. Dew harvesting depends on atmospheric vapor condensing on a sufficiently favorable surface, making temperature and dew point important.

Sorption-based AWH uses hygroscopic materials to adsorb or absorb moisture and later release it during regeneration.

These mechanisms should not be mixed when discussing performance. A fog collector can be evaluated through interception and liquid drainage, while a dew system depends strongly on thermal conditions and a sorbent system involves adsorption/desorption behavior. The present research focuses mainly on passive fog and surface/interface design.

2.3 Passive versus active AWH

Active AWH can use electricity, refrigeration, thermoelectric cooling, or other powered processes. Such systems can potentially reach higher controlled yields, but their energy demand can be a major barrier. Passive AWH relies on natural or sustainable driving forces such as airflow, gravity, capillary action, curvature effects, surface-energy gradients, or radiative cooling.

2.4 The role of surface engineering

Liu et al. identify three broad passive enhancement strategies: surface/morphology optimization, substrate cooling, and sorbent assistance. Surface optimization primarily addresses interfacial dynamics-how water nucleates, grows, coalesces, moves, and leaves. Cooling changes the thermal condition for condensation, while sorbents increase local moisture capture. The present research concentrates on the surface/interface component while recognizing that temperature and humidity remain boundary conditions.

This framework makes the research question more rigorous. Rather than asking simply, "What surface collects the most water?", the engineering question becomes: How can surface morphology and wettability be optimized to improve droplet capture, coalescence, directional transport, and collection under defined atmospheric conditions?

3. Atmospheric Water Collection as a Multistage Process

3.1 Capture and interception A passive fog collector must first encounter water. Fog-laden air contains suspended droplets, and the collector needs sufficient exposed area and geometry to intercept them. Dong et al. (2025) discuss fog droplets on the approximate scale of 1-40 μm and emphasize the importance of droplet-size distribution and airflow. Consequently, fog harvesting is partly an aerodynamic problem as well as a surface-chemistry problem.

Cactus-inspired structures demonstrate why geometry matters. A tapered three-dimensional spine presents a structured obstacle to airflow rather than a completely flat plane. At larger scales, mesh opening, fiber arrangement, collector orientation, and overall geometry can influence how much moisture reaches the active surface.

3.2 Nucleation and initial water interaction After water reaches the surface, wetting behavior becomes important. Hydrophilic surfaces generally favor stronger interaction and spreading, while hydrophobic surfaces generally favor beading and reduced wetting. However, the ten-source literature does not support a universal claim that one behavior is always superior.

Zhang and Guo (2020) describe the tradeoff clearly: hydrophilic surfaces can promote deposition and capture, but excessive spreading can create films; hydrophobic surfaces can favor droplet growth and removal, but initial deposition can be slower.

Patterned wettability can therefore assign different functions to different regions.

3.3 Droplet growth and coalescence Captured droplets can grow through condensation, additional fog-droplet impact, and coalescence with neighboring droplets. Coalescence is important because water must eventually become mobile enough to leave the active collection region. Pan et al. 2026) organize biomimetic collection around nucleation, consolidation/coalescence, directional transport, and storage/collection.

This sequence creates a design requirement: the surface needs enough residence time for droplets to consolidate, but not so much adhesion that large droplets become permanently pinned. Thus, droplet residence time and contact-angle hysteresis become engineering variables.

3.4 Directional transport, the reviewed literature identifies several possible transport mechanisms: curvature-induced Laplace-pressure gradients, surface-energy or wettability gradients, capillary forces, and gravity. These mechanisms can operate together instead of independently.

Nawaz et al. (2026) describe cactus spines as tapered cones whose changing curvature creates a Laplace-pressure bias.

Longitudinal microgrooves provide local capillary guidance. A spatial wettability gradient can reinforce the direction, while gravity can assist after droplets become larger. The result is a hierarchical transport pathway in which microgeometry establishes direction and microgeometry maintains it.

3.5 Collection and surface renewal, the transport pathway must eventually deliver water to a defined outlet or reservoir. This is part of the harvesting mechanism rather than an afterthought. Removing water from the active region exposes fresh surface for additional droplets. If water remains pinned, the collector can become progressively covered and less effective.

FUNCTIONAL CYCLE Atmospheric moisture → interception/capture → nucleation or retention → droplet growth → coalescence → directional transport → collection → surface renewal → repeat.

4. Wettability, Surface Chemistry, and Droplet Mobility

- 4.1 Young's equation Wettability is influenced by surface chemistry as well as structure. Al-Gunaid et al. (2025) review Young's equation for an ideal smooth surface:
  cos(theta_c) = (gamma_SG - gamma_SL) / gamma_LG
  where theta_c is the equilibrium contact angle and the gamma terms represent interfacial tensions. The equation provides a conceptual connection between surface chemistry and wetting. Real biomimetic surfaces, however, are often rough, patterned, chemically heterogeneous, and hierarchical, so the ideal model is only a starting point.
- 4.2 Wenzel roughness for a rough chemically homogeneous surface, the Wenzel model is commonly written as:
  cos(theta_W) = r cos(theta_c)

where r is the roughness ratio. Roughness changes the effective wetting response. This is important because many biological surfaces achieve their behavior through micro- and nanoscale texture rather than chemistry alone.

4.3 Cassie Baxter behavior In a Cassie-type state, a droplet can contact a combination of solid and trapped air. This can reduce effective solid-liquid contact and support low adhesion. Lotus leaves are a classic example. Wan et al. (2021) report literature values of approximately 164° water contact angles and a slide angle near 3° for lotus-inspired super hydrophobicity. However, extremely low adhesion should not automatically be treated as optimal for water harvesting. Rapid shedding can help refresh the surface, but excessive repellency can also reduce residence time for initial capture. The engineering target is therefore not maximum hydrophobicity; it is the correct combination of capture and mobility.

4.4 Contact-angle hysteresis and pinning Contact angle alone does not determine whether a droplet will move. Contact-angle hysteresis, associated with advancing and receding contact angles, provides information about pinning. Higher hysteresis
generally, means stronger resistance to motion. Lower hysteresis tends to support mobility. Al-Gunaid et al. (2025) identify contact-angle hysteresis as important for directional transport and discuss three major driving contributions: chemical/wettability gradients, structural or Laplace-pressure gradients, and gravity. The source notes express a
conceptual gravity-versus-pinning comparison as Fg = rho g R^3 and Fp = gamma_w(cos theta_r - cos theta_a), with transport favored when driving forces exceed resistance. The exact balance depends on geometry and assumptions.

4.5 Patterned wettability as functional zoning The Namib Desert beetle demonstrates why a collector can benefit from spatial rather than uniform wettability. Hydrophilic regions provide preferential water-interaction sites, while neighboring hydrophobic regions can facilitate movement. Zhang and Guo and Al-Gunaid et al. both use this concept to explain biomimetic patterned surfaces.

The engineering implication is a shift from searching for one ideal contact angle to designing a sequence of functions: a capture zone, a coalescence zone, a low-adhesion transport path, and a hydrophilic drainage channel. This functional zoning is one of the central design principles carried forward into the proposed collector.

5. Biomimetic Design Principles and Engineering Significance

5.1 Namib Desert beetle: spatially separated functions The Namib Desert beetle is one of the most influential biological models in the reviewed literature. Its dorsal surface is described as containing hydrophilic protuberances surrounded by comparatively hydrophobic wax-coated regions. Al-Gunaid et al. 2025) discuss hydrophilic bumps of approximately 0.5mm diameter and representative spacing of roughly 0.5-1.5mm in the reviewed literature. These biological dimensions are reference observations rather than fixed engineering requirements.

The important lesson is heterogeneous wettability. Water can preferentially interact with selected areas while neighboring areas provide a lower-adhesion environment for movement. This offers a natural solution to the capture-versus-transport conflict.

5.2 Cactus spine: geometry as a passive transport mechanism Cactus spines provide a complementary solution. Nawaz et al.

(2026) describes a narrow apex and broader base, with curvature changing along the spine. This creates a spatially varying interfacial pressure environment associated with a Laplace-pressure gradient. Longitudinal microgrooves add local capillary guidance and can constrain droplet motion.

The cactus mechanism therefore works across length scales. The macro-scale taper establishes direction, the microgrooves guide water locally, surface-energy differences can reinforce transport, and gravity can assist after water grows larger. The source notes also discuss a cited theoretical result in which a barrel-shaped droplet below approximately 440 uL could move along cactus-like microgrooves; this is a reported theoretical result, not a universal threshold.

5.3 Spider silk: periodic accumulation and redistribution Spider silk contains repeated spindle-like regions and junctions that can act as preferred sites for droplet accumulation. Zhang and Guo (2020) and Wan et al. (2021) describe how curvature and surface-energy differences can contribute to droplet movement along engineered spider-silk-like fibers.

For an engineering collector, this architecture is particularly relevant to mesh or fiber systems. Periodic nodes could provide localized coalescence points before water enters a larger drainage pathway. However, extra structure should only be included when its function justifies additional manufacturing complexity or fouling risk.

5.4 Mango leaf: hierarchical drainage Foday Jr. et al. (2022) provides an experimental example of biological morphology being transferred into an engineered polymer. A Mangifera indica leaf was replicated in PDMS while preserving primary and secondary veins, curvature, microgrooves, and an apex. The study used SEM, FTIR, contact-angle measurements, optical imaging, and video analysis.

Under the study’s-controlled conditions-approximately 21 + 3 °C, 80-85% RH, 45° inclination, and humidifier-generated moist airflow the replicated mango-leaf surface reportedly collected about 7.02 g cm h, compared with approximately 6.20 g cm•2 h' for the dry natural leaf and 5.89 g cm? h' for the fresh leaf. These are published experimental values from Foday Jr. et al.; they are not results obtained in the present research.

The transferable finding is architectural. The leaf's veins and microgrooves form a hierarchy in which many local pathways converge toward larger drainage routes and a defined apex. This

suggests that a biomimetic collector should not simply maximize active surface area; it should also provide a controlled route for moving water to a reservoir.

5.5 Integrated engineering interpretation the biological models solve different portions of the same engineering problem. The beetle provides where to capture. The cactus provides how to move water directionally. Spider silk provides how to accumulate and redistribute droplets at repeated nodes. Mango leaves provide how to merge many small transport paths into a larger drainage system. Shorebird-beak inspiration adds a capillary mechanism for directional movement.

The resulting preliminary architecture for the later research is a passive collector containing repeated cactus-like tapered units, beetle-inspired wettability zoning, longitudinal microgrooves, mango-leaf-inspired hierarchical drainage, and a final reservoir.

6.1 Why the sources are complementary

The ten sources do not all answer the same question. Several are broad reviews that establish the physics and biomimetic design vocabulary, while Sources 8 and 10 provide experimental demonstrations. Together, they support a system-level interpretation: an atmospheric-water collector must coordinate water capture, droplet growth/coalescence, directional transport, collection/storage, and surface renewal rather than optimizing only one property.

6.2 Source-by-source contribution

- Source 1- Zhang & Guo (2020): establishes the capture → transportation → storage framework and explains microstructure, wettability, surface-energy gradients, Laplace pressure, capillarity, and gravity.
- Source 2 - Nawaz et al. (2026): concentrates on cactus-inspired fog harvesting; tapered geometry, grooves, curvature, Laplace pressure, wettability gradients, capillary forces, and gravity are central.
- Source 3 - Al-Gunaid et al. (2025): focuses on engineered surfaces and patterned wettability, including contact-angle theory, hysteresis, transport forces, fabrication, and measured examples.
- Source 4 - Liu, Beysens & Bourouina (2022): places surface engineering inside passive AWH and distinguishes morphology/wettability from cooling and sorbent assistance.
- Source 5 - Pan et al. (2026): expands the biological design library to spider silk, desert beetle, cactus spine, and shorebird beak and frames collection as nucleation → consolidation/coalescence → transport → storage.
- Source 6 - Dong et al. (2025): emphasizes fog-droplet behavior, aerodynamics, surface-energy/Laplace-pressure transport, multiscale structures, fouling, durability, and standardized testing.
- Source 7 - Wan, Gou & Guo (2021): identifies six biomimetic mechanisms: wettability, transmission area, long-distance delivery, accumulation/storage, condensation promotion, and gravity-driven collection.
- Source 8 - Foday Jr. et al. (2022): experimentally studies a Mangifera indica leaf bio template, showing how morphology, wettability, inclination, veins, grooves, and droplet dynamics interact.
- Source 9 - Tao et al. (2026): connects AWH with solar-driven interfacial evaporation and identifies photothermal management, directional transport, atmospheric capture, and interface stability. Evaporation results must not be presented as fog/AWH collection rates.
- Source 10 - Choi, Baek & So (2023): directly combines cactus-inspired hierarchical geometry with Namib-beetle-inspired dual wettability using 3D-printing-assisted fabrication.

6.3 Cross-source pattern

Repeated conclusion: wettability alone is insufficient. Hydrophilic regions can promote deposition or transfer, while hydrophobic/low-adhesion regions can facilitate droplet growth and

removal. Spatially patterned wettability allows different regions to perform different functions. Hierarchical geometry provides physical pathways for droplet motion.
The strongest synthesis is therefore a division of labor: capture - coalescence - transport → collection, with geometry and surface chemistry designed together.

7.1 Why reported numbers must be interpreted carefully
The sources use different units, humidity levels, airflow conditions, surface areas, collection times, and testing configurations.
Therefore, reported rates should be used as evidence that a design can work under a particular test condition —not as a direct ranking of all materials. The reviews also identify the lack of standardized evaluation as a barrier to fair comparison.
7.2 Important experimental evidence

Source	System/finding	Reported result	Condition/interpretation
3	Engineered PDMS Plasma-treated surface	1.14±0.06 g/cm^2 h	Best cited example; pristine PDMS 0.14±0.01.
3	PAN/EG electro spun	774 mg/cm^{2}h	Reported as 91% higher than virgin PAN in the cited study.
3	Orientation optimization	45º best	Humidifier distance also mattered;>5cm reduced efficiency in cited study
8	Replicated mango leaf	7.02 g/cm^2 h	21±3ºC, RH 80-85%, 45º inclination, humidifier perpendicular.
8	Dry mango leaf	6.20 g/cm^2 h	Same controlled study conditions.
8	Fresh mango leaf	5.89 g/cm^2 h	Same controlled study conditions.
10	Best dual-wettability hierarchical surface (BI)	7.85 g average weight/10min	Specific controlled fog experiment; not a universal collection rate.
1	Cactus/spiderweb-inspired examples	0.09 and 0.8 mL/min	Review-reported examples; not directly comparable across studies.

7.3 The most useful lesson from the numbers
Source 10 is particularly relevant because it tests flat versus hierarchical and hydrophobic versus hydrophilic configurations. Its conclusion supports a combined strategy: the best-performing design used hierarchical geometry plus a hydrophilic base/bottom and hydrophobic tip. Source 8 similarly shows that a replicated natural morphology can outperform the natural leaf in its controlled setup, illustrating how geometry and surface state can be engineered together.
Do not write: "Hydrophobic surfaces always collect more water." Write instead: "Hydrophobicity, hydrophilicity, contact-angle hysteresis, geometry, inclination, and droplet size interact; the optimal configuration depends on the collection/transport stage and test environment."

Research Gap and Problem Definition

8.1 What the literature has established

- Biological surfaces organize water using combinations of surface chemistry, roughness, curvature, grooves, fibers, and gradients.
- Namib beetles demonstrate heterogeneous wettability; cactus spines demonstrate tapered/grooved transport; spider silk demonstrates periodic accumulation/transport; mango leaves demonstrate vein/groove-guided drainage.
- Engineered surfaces can reproduce or combine these mechanisms through plasma treatment, coatings, lithography, electrospinning, templating, and 3D printing.
- Droplet mobility is affected by contact-angle hysteresis, adhesion/pinning, surface-energy differences, Laplace pressure, capillarity, gravity, and geometry.

8.2 What remains unresolved

1. Durability: UV exposure, dust/sand, salt, fouling, wet-dry cycling, mechanical damage, and degradation of coatings or lubricants remain practical concerns.
2. Scalable fabrication: High-resolution micro/nanostructures can be expensive or difficult to reproduce over large areas.

Small-coupon performance does not automatically imply manufacturability at collector scale.

1. Standardized testing: Different RH, fog concentration, airflow, temperature, droplet distributions, collection times, and measurement methods make comparisons difficult. Controlled chambers may not represent outdoor environments.
2. Integrated multistage design: Many studies emphasize one mechanism-capture, condensation, transport, or collection. The engineering challenge is connecting these mechanisms without creating a bottleneck at the next stage.
3. Environmental robustness and water quality: A collector must tolerate dust, biological contamination, salts, changing weather, and repeated wet/dry cycles. Collection quantity alone does not establish drinking-water safety.

- 8.3 Research gap statement for the poster
  "Although biomimetic surfaces have demonstrated improved atmospheric water capture and transport under controlled conditions, a persistent gap remains in integrating multiscale geometry and spatially patterned wettability into a durable, scalable, and environmentally robust passive collector that can be evaluated using standardized, field-relevant conditions."
- 8.4 Why this gap fits the project

The project does not need to solve the entire commercialization problem. Its defensible contribution is to synthesize design rules, identify where mechanisms complement or conflict, and propose a conceptual architecture whose components have traceable biological and physical rationales.

Proposed Conceptual Biomimetic Collector

9.1 Design objective

Design a passive, modular collector that separates major functions across a hierarchical surface: capture incoming moisture/droplets, encourage coalescence, move water directionally, and deliver it to a protected collection channel.

9.2 Proposed architecture

Component	Biological inspiration	Engineering function
Open macro support / mesh	Spiderweb / fog-catching structures	Intercept moving fog while maintaining airflow.
Tapered conical units	Cactus spines	Create curvature/pressure differences favoring directional movement.
Microgrooves	Cactus surface grooves	Provide drainage paths and increase effective area.
Hydrophobic/low-adhesion capture region	Namib beetle background / engineered surfaces	Allow droplet growth and reduce excessive pinning where appropriate.
Hydrophilic base / transport region	Namib beetle patches / plant transport zones	Promote transfer toward downstream collection.
Hierarchical vein/channel network	Mango leaf	Collect water from microstructures and direct it to an outlet.
Optional periodic nodes	Spider silk spindle-knot architecture	Provide localized accumulation/coalescence sites if needed.
Protected outlet/reservoir	Biological storage analog	Receive water after directional transport.

9.3 Mechanism chain

Fog/moisture arrives → interception → droplet deposition/nucleation → growth → coalescence → transition to an appropriate wetting region → Laplace/capillary/gravity-assisted movement along taper and grooves → hierarchical channel collection → reservoir.

The key design principle is spatially separated functions. The surface does not need one wettability value everywhere; each region is assigned a role based on whether the priority is capture, growth, mobility, or collection.

9.4 Parameters for future optimization

- Cone/spine height, tip radius, base radius, and taper angle.
- Microgroove width, depth, spacing, and orientation.
- Spacing and areal density of collector units.

- Hydrophobic-to-hydrophilic pattern size and spacing.
- Contact angle plus advancing/receding contact angles or contact-angle hysteresis.
- Collector inclination relative to gravity.
- Fog droplet size distribution, airflow velocity, temperature, and relative humidity.
- Durability after repeated wet/dry, dust, UV, and mechanical exposure.

Literature-Based Methodology, Engineering Significance &
Final Synthesis

10.1 Literature-based methodology

This project can be described as a structured literature synthesis followed by conceptual engineering design. The ten selected peer-reviewed sources are analyzed using a common extraction framework so biological observations can be translated into engineering requirements.

Workflow

1. Identify biological prototype	Record the organism/structure and its water-harvesting behavior.
2. Extract surface architecture	Record macro-, micro-, and nanoscale geometry: cones, grooves, fibers, veins, bumps, nodes, channels, and roughness.
3. Extract wetting behavior	Record hydrophilic, hydrophobic, superhydrophobic, biphilic/dual-wettability, gradients, contact angle, and hysteresis where available.
4. Identify physical mechanism	Classify deposition/capture, condensation, coalescence, Laplace-pressure motion, capillarity, surface-energy gradient, gravity, or combinations.
5. Record fabrication route	Compare templating, coating, plasma treatment, electrospinning, lithography, 3D printing, and other approaches.
6. Record performance with conditions	Keep every reported value tied to RH, temperature, airflow/fog source, inclination, time, area, and measurement method when provided.
7. Extract limitations	Track durability, fouling, environmental stability, scalability, cost, water quality, and testing-standard limitations.
8. Convert evidence into design requirements	Use recurring mechanisms to justify each component of the proposed collector.
9. Propose conceptual architecture	Combine compatible mechanisms while explicitly labeling the design as conceptual and not experimentally validated

10. Define future validation	Specify variables and controlled tests needed to determine whether the architecture improves collection.

10.2 Engineering significance

The project translates biological observations into engineering design variables. That makes it relevant to mechanical engineering because the problem involves geometry, manufacturing, fluid behavior, surface interactions, parameter optimization, testing, and system integration. The proposed architecture can later be modeled in CAD and fabricated through additive manufacturing or molding, but those steps belong to future validation rather than the present literature study.

- 10.3 Final synthesis / thesis
  The literature indicates that effective passive atmospheric water harvesting is best understood as a coordinated interfacial transport problem rather than a simple hydrophobic-versus-hydrophilic choice. Biomimetic systems demonstrate that hierarchical geometry can control where droplets form and move, while patterned wettability can assign different regions to capture, retention, transfer, and release. A cactus-inspired tapered and grooved structure, combined with beetle-inspired heterogeneous wettability and a mango-leaf-inspired drainage network, therefore provides a rational conceptual pathway for integrating capture and directional transport into one passive collector.
- 10.4 What the project can legitimately claim
- The literature supports specific physical mechanisms and design principles.
- Published experiments demonstrate engineered biomimetic surfaces can improve water collection under their reported conditions.
- The proposed collector is a literature-derived conceptual design.
- The project identifies measurable parameters for future optimization.

It cannot claim yet: a new collection rate, experimentally proven superiority of the proposed design, long-term outdoor durability, drinking-water safety, or commercial viability. Those require experiments and/or field validation.

Design Requirements Derived from the Literature

11.1 From biomimicry to engineering requirements

A useful biomimetic engineering process does not simply copy an organism. It identifies the biological function, determines the physical mechanism responsible for that function, and translates it into a controllable engineering variable. Across the ten sources, the recurring functions are capture, droplet growth/coalescence, directional transport, collection, and surface renewal.

Observation from literature	Engineering requirement	Variables to measure
Namib beetle: hydrophilic regions on hydrophobic background	Use spatially patterned wettability	Contact angle; advancing/receding angles; pattern size and spacing.
Cactus: tapered spine + grooves	Create a converging, guided transport path.	Taper angle; tip/base radius; groove dimensions.
Cactus: curvature/Laplace pressure	Use changing curvature to encourage directional motion.	Droplet curvature; pressure difference;
Mango leaf: veins + grooves + apex	Provide hierarchical downstream drainage.	Channel dimensions; branch spacing; outlet geometry.
Spider silk: periodic spindle/knot regions	Consider localized coalescence sites if they provide a benefit.	Node size; spacing; droplet residence time.
Fog-harvesting reviews: airflow is important	Keep the collector open enough for fog interception.	Open-area ratio; airflow velocity; pressure drop.
Reviews: dust, UV, salt and wet/dry cycles are concerns	Preserve geometry and wetting behavior after exposure.	Wetting-state change; mass loss; performance retention.

11.2 Core design constraints

- Passive operation: the basic collector should not depend on active refrigeration, pumps, or electrically driven airflow.
- Gravity-assisted drainage: orientation should help move water toward the outlet.
- Airflow compatibility: fog must reach the active surface without excessive blockage.
- Modularity: repeated units make the design easier to manufacture, compare, replace, and scale.
- Maintainability: contamination should not permanently destroy the intended surface behavior.

11.3 Central design rule

Do not optimize a single surface property in isolation. The literature supports coordinated optimization of geometry + wettability + orientation + environmental conditions. The proposed collector should therefore be treated as a coupled surface-fluid system.

Physics of Droplet Capture, Motion, and Removal

12.1 Wettability

The literature uses contact angles as a central descriptor of wetting. Young's equation relates the equilibrium contact angle to solid-gas, solid-liquid, and liquid-gas interfacial energies. Surface roughness modifies apparent wetting, represented by the Wenzel framework, while Cassie-Baxter behavior describes composite interfaces involving trapped gas.

For this project, the important point is functional rather than categorical. Hydrophilic regions can promote interaction and transfer, while hydrophobic or low-adhesion regions can facilitate droplet growth and release in suitable designs. Several sources therefore support spatially patterned rather than uniformly optimized wetting.

12.2 Contact-angle hysteresis and pinning

Advancing and receding contact angles provide information about droplet mobility. A large difference between them generally indicates stronger pinning/hysteresis. Consequently, a surface should not be evaluated only by one static contact angle when directional transport is the goal.

12.3 Laplace pressure

Curvature-induced Laplace pressure is repeatedly identified as a transport mechanism. A useful conceptual relation is AP = y(1/R• + 1/R•), where y is liquid surface tension and R and R are principal radii of curvature. Tapered cactus-like structures can create changing curvature conditions that contribute to directional motion.

Important: this equation is a physical framework from the literature; the current project does not calculate a new pressure field for its proposed collector.

12.4 Surface-energy gradients

A transition between different wetting states can encourage droplet relocation. Beetle-inspired systems use hydrophilic capture regions surrounded by hydrophobic material, while engineered dual-wettability surfaces can deliberately place different wetting states at different positions. Source 10 is particularly important because it combines this effect with cactus-inspired geometry.

12.5 Gravity and capillary forces

Gravity can assist drainage when the component of droplet weight exceeds resisting adhesion/pinning. Capillary forces can also pull liquid through narrow grooves and channels. The proposed transport path should allow surface forces to initiate movement and gravity to assist the downstream flow.

12.6 Surface refresh

Droplet removal is more than a final collection step. Removing water exposes fresh active surface, allowing subsequent droplets to arrive and grow. Efficient transport can therefore improve the continued availability of the capture surface.

12.7 Engineering interpretation

Capture efficiency and transport efficiency are different quantities. A surface may intercept many droplets but retain them, while another may capture fewer droplets but rapidly release

them. The proposed design seeks a sequence in which each stage prepares the surface for the next.

Proposed CAD / Concept Development Framework

13.1 How the conceptual collector should be represented

The collector should be modeled as a modular repeating surface rather than one unnecessarily complicated biomimetic object. This makes the design easier to explain, manufacture, modify, and eventually test. A single unit can be developed first, followed by an array and then a manifold drainage.

13.2 CAD hierarchy

Level CAD feature	Purpose
Support frame / open mesh	Structural support and exposure to moving fog.
Tapered collector unit	Primary cactus-inspired geometry.
Longitudinal microgrooves	Directional guidance and additional structured area.
Wettability regions	Hydrophobic/low-adhesion capture zone + hydrophilic transport zone.
Branching drainage network	Mango-leaf-inspired collection pathway.
Manifold/outlet	Combines flow from repeated units into one protected stream

13.3 Geometry variables

- Taper angle, tip radius, and base radius.
- Groove width, depth, spacing, and orientation.
- Spacing and areal density of repeating units.
- Array inclination is related to gravity.
- Drainage-channel dimensions and outlet geometry.

13.4 Wettability pattern variables

- Hydrophobic tip / hydrophilic base is directly supported by Source 10.
- Uniform hydrophobic and uniform hydrophilic surfaces can serve as conceptual controls.
- Pattern dimensions should be treated as optimization variables, not assumed universal constants.

Future Experimental Validation Plan

14.1 Purpose of validation

The literature synthesis generates testable hypotheses but does not validate the proposed collector. A future experiment should isolate variables so that improvements can be attributed to geometry, wettability, or their interaction.

14.2 Recommended comparison set

Sample Geometry	Wettability	Purpose
Flat	Uniform hydrophobic	Baseline for low-adhesion behavior.
Flat	Uniform hydrophilic	Baseline for wetting/capture behavior.
Hierarchical	Uniform hydrophobic	Tests geometry without patterned wettability.
Hierarchical	Uniform hydrophilic	Tests geometry with strong wetting.
Hierarchical	Hydrophobic tip / hydrophilic base	Tests proposed cactus + beetle synergy.
Hierarchical + drainage network	Hydrophobic tip / hydrophilic base	Tests full conceptual architecture.

14.3 Variables to control

- Relative humidity and temperature.
- Fog/moisture generation rate and droplet-size distribution.
- Airflow velocity and direction.
- Surface inclination.
- Collection area and test duration.
- Initial surface condition and cleaning procedure.

14.4 Performance metrics

- Water collection rate: mass or volume collected per unit area per unit time.
- Droplet transport time: time from capture region to collection region.
- Removal/refresh behavior: how quickly droplets leave the active surface.
- Contact-angle hysteresis: advancing versus receding angle as a mobility indicator.
- Durability: change in wetting state and collection performance after repeated exposure/cycles.

14.5 Future hypothesis

Hypothesis: A hierarchical surface combining cactus-inspired tapered/grooved geometry with beetle-inspired hydrophobic-tip/hydrophilic-base patterning will provide more effective directional droplet transport than comparable flat or uniformly wetted surfaces under the same controlled fog conditions.
This is explicitly a future hypothesis, not a result of the present literature study.

Final Research Synthesis and Paper/Poster Blueprint

15.1 The complete argument

The literature supports a coherent engineering story. Atmospheric water occurs as vapor or suspended droplets, and passive collection depends strongly on relative humidity, temperature, dew point, airflow, and fog characteristics. At the surface, geometry and wettability govern interaction with water. Biological systems demonstrate that these properties can be spatially organized: beetles concentrate capture through heterogeneous wettability, cacti use tapered/grooved structures for directional transport, spider silk uses periodic structures for droplet accumulation and movement, and leaves use hierarchical veins and grooves for drainage. The unresolved engineering challenge is not whether individual biological mechanisms can work. It is how to integrate complementary mechanisms into a scalable, durable, field-relevant collector while preserving each stage's function and evaluating performance under comparable conditions.

15.2 Poster blueprint

- Top: title + research question + objective.
- Left: freshwater problem + atmospheric water collection process.
- Center: large biological-to-engineering diagram: beetle → cactus → mango leaf → proposed collector.
- Right: literature comparison + selected published experimental evidence.
- Bottom: research gap + conceptual design + future validation + conclusion.

15.3 Conclusion

The reviewed literature suggests that passive atmospheric water harvesting can be advanced by coordinating surface wettability with hierarchical geometry rather than relying on a single hydrophobic or hydrophilic state. Beetle-inspired heterogeneous wettability can support functional separation between capture and release, cactus-inspired taper and grooves can promote directional transport, and leaf-inspired hierarchical channels can organize downstream collection. The resulting conceptual collector provides a literature-grounded engineering framework for future fabrication and controlled validation.

Synthesis of the Literature

16.1 What the ten sources collectively show

Across the reviewed literature, atmospheric water harvesting emerges as a multistage interfacial transport problem. Water must first interact with a collector, then form or accumulate into droplets, coalesce when appropriate, move toward a collection region, and finally be removed from the active surface. The sources differ in biological model and fabrication method, but they repeatedly return to the same physical variables: surface chemistry, roughness, curvature, wettability, contact-angle hysteresis, surface-energy differences, capillary forces, Laplace pressure, gravity, airflow, temperature, and relative humidity.

16.2 The central finding

The literature does not support a simple conclusion that either hydrophobicity or hydrophilicity is universally superior. Instead, the evidence supports assigning different surface states to different functions. Hydrophilic regions can facilitate interaction and transfer, whereas hydrophobic or low-adhesion regions can support droplet growth and release under appropriate conditions. Patterned wettability can therefore create a functional division of labor across the collector.

16.3 Why hierarchical geometry matters

Geometry determines more than surface area. Cactus-inspired tapered structures provide curvature and a directional pathway; microgrooves can guide water; leaf-inspired veins provide hierarchical drainage; spider-silk-like periodic structures can provide localized accumulation sites. These structures interact with wetting behavior rather than operating independently.

16.4 Why the combination is important

Source 10 provides especially direct support for combining cacti-like hierarchical structure with beetle-inspired dual wettability. Source 8 provides complementary evidence that natural leaf morphology can be replicated and that the resulting surface can display different collection behavior under controlled humidifier conditions. The reviews broaden these findings and identify durability, scalability, fouling, environmental conditions, and testing standardization as unresolved challenges.

16.5 Synthesis statement

Therefore, the strongest literature-supported design principle is coordinate surface wettability and hierarchical geometry so that capture, droplet growth, directional transport, and collection are treated as connected but distinct functions.